ABHISHEK SHRIVASTAVA

Senior Data Engineer | Multi-Cloud Data Platforms (AWS · Azure · Snowflake · Hadoop) | Production AI/ML & Agentic Systems

Event-Driven & Metadata-Driven Pipelines · Lakehouse & Warehouse Architecture · Agentic & RAG/LLM Systems · Governance & Observability

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PROFESSIONAL SUMMARY

Senior Data Engineer with 7 years building high-scale, production data platforms across AWS, Azure, Snowflake, and Hadoop — event-driven and metadata-driven pipelines, lakehouse and warehouse architectures, dimensional modeling, and the data-quality, governance, and observability that keep them trustworthy. Proven across multiple clouds and paradigms: Spark/PySpark/Snowpark, dbt, Airflow, Iceberg lakehouse patterns, and both greenfield builds and ownership of inherited legacy systems. Track record of cutting pipeline latency from hours to minutes, scaling to billions of events/day, and mentoring engineers across teams. Now extends that same platform rigor into production AI/ML — RAG systems built with evaluation, grounding, and observability rather than guesswork.

SKILLS

Data Engineering: Event-driven & metadata-driven architecture, ETL/ELT, dimensional modeling (star schema, SCDs), lakehouse (Iceberg), data warehousing, streaming, data quality, governance, lineage & auditability

Databases & SQL: Expert SQL (window functions, CTEs, query optimization), Snowflake (performance tuning, modeling), Redshift, PostgreSQL, Hive, SQL Server, DB2

Cloud & Tooling: AWS (Glue, S3, Lambda, EventBridge, Redshift, EMR, DMS), Azure DevOps, dbt, Airflow, Autosys, CI/CD, Docker/Kubernetes, GitHub Actions, Nexus, FastAPI

Big Data & Languages: Spark, PySpark, Snowpark, Iceberg, Kafka, Hadoop (Cloudera/Hortonworks), distributed systems · Python, SQL

AI/ML: Production RAG, agentic workflows, hybrid retrieval, cross-encoder reranking, embeddings (sentence-transformers), RAG evaluation (recall@k, MRR), LLM integration, pgvector, PyTorch

PROFESSIONAL EXPERIENCE

Senior Data Engineer — EDP Platform, MUFG, Bengaluru

Jan 2024 – Present

AWS (Glue · EventBridge · Iceberg · Redshift · Lambda · DMS) · Spark · Python · SQL · Azure DevOps · Autosys · CI/CD · Agentic Tooling (Copilot-driven)

- Designed and led a **modern, event-driven data architecture on AWS** (DMS → S3 → EventBridge → Glue → Iceberg → Redshift) powering near real-time model-risk reporting across multiple risk domains.
- Built **metadata-driven orchestration** (Postgres job control governing stage → transform → enterprise → info-mart stages) that generates pipelines from config.
- Independently own and maintain a separate **legacy metadata-driven architecture** — a Postgres-cataloged lineage system with Autosys-scheduled Hive transformations — carrying 8-9 additional risk pipelines with full raw-to-enterprise lineage and auditability on a previously undocumented, inherited system.
- Built **Plan Craft**, an **agentic + deterministic** query-optimization web app spanning **Hive, SQL Server, DB2, and Snowflake** — an agentic workflow profiles each DB's tables/stored procedures/queries for runtime evidence, applies deterministic per-DB optimization rules, then has an LLM generate two candidate rewrites; all versions are benchmarked live against the original to prove the optimization before returning results, with per-DB context profiles, batch SQL input, and history. Cut query-tuning time to minutes and improved query performance by **~8-28%** for **15-20 engineers** across three platform teams (EDP, EMR, DB2).
- Built **CI/CD on Azure DevOps** for the Glue deployment pipeline — a gateway distributes the full ~172-script package that powers the Glue jobs, sourced from a Nexus repo, with each script environment-prefixed for safe, consistent promotion across Dev, UAT, Prod, and DR.
- Cut pipeline latency from **hours to minutes** by standardizing validations — row-hash reconciliation, schema-evolution guards, and versioned-partition reproducibility for audit — and migrated historical datasets from Redshift to an **Iceberg lakehouse** for time-travel and cost-efficient retrieval.
- Mentored and onboarded an external EY consultant onto the platform in roughly a week, enabling independent contribution to ongoing maintenance.

Data Engineer — Data Intelligence, SailPoint Technologies, US Remote / Pune — Mar 2022 – Jan 2024

Snowflake · dbt · Airflow · Python · Spark / Snowpark · AWS · Kafka · CI/CD

- Designed **Snowflake data models** and **Kafka-based streaming ingestion** powering IdentityAI, processing billions of events/day; built dimensional models and governance patterns for analytics at scale.
- Built a **"Pipeline Factory"** to bootstrap new Airflow DAGs from config files — reduced onboarding time ~70% and standardized data-quality and governance patterns across teams.
- Drove cost optimization by replacing Flink with **Spark-based** Jaccard computation; multi-tenant design over billions of rows/day, saving in the order of millions annually.
- Led a **Snowpark + dbt** POC centralizing testing and data-quality checks; built CI/CD and migrated internal packages to AWS CodeArtifact for versioned distribution.
- Owned on-call — added alerting/observability, triaged incidents, and coordinated infra fixes for high SLA adherence.

Deputy Manager — Data Science & Engineering, DCB Bank, Mumbai — Oct 2019 – Mar 2022

Hadoop (Cloudera/Hortonworks) · Hive · Pentaho · Python · SQL

- Engineered **end-to-end pipelines** for Internet/Mobile Banking; migrated workloads to Hadoop/Hive for scalable reporting and lower TCO.
- Automated **RBI compliance** & operational reporting (financial inclusion, NEFT/RTGS, Video-KYC, ATM dispensations, gold-loan exposure).
- Delivered daily MIS & executive dashboards with validation and freshness/quality alerting; **mentored 6+ engineers** across the Big Data stack.

SELECTED PROJECTS

AI Incident Copilot — Production RAG for incident root-cause analysis

- Built a production RAG pipeline: hybrid retrieval (SQL time-filter + pgvector dense search), cross-encoder reranking, and a confidence quality-gate that suppresses low-grounding answers. FastAPI, Dockerized, CI/CD, 267 tests.
- Built a 31-question evaluation harness (recall@k, MRR) — measured perfect lookup retrieval (recall@5 = 1.0) and surfaced a multi-hop gap (recall@5 = 0.33), diagnosed to single-pass retrieval with scoped fixes.
- Chose a deterministic pipeline over an agent for the core flow — incident triage has known steps, and trust under pressure beat autonomous cleverness.

WRITING

Building a Modern Event-Driven Data Architecture on AWS — Medium

EDUCATION & CERTIFICATIONS

- AWS Certified Cloud Practitioner (2023)
- Springboard — Data Science Program (2021)
- PG Diploma, Big Data Analytics — CDAC (2019)
- B.E. Computer Science — RGPV (2018)